EXPERIMENT – 1

BASIC CELLULAR SYSTEM

Aim

To design and illustrate a basic cellular communication system using Figma, including its architecture, cell layout, frequency reuse, handover process, key components, and a simple prototype.

Procedure

1. Open Figma and create a new design file.

2. Create separate frames for the cellular architecture, cell layout, frequency reuse, handover, and key components.

3. Use shapes, icons, labels, and arrows to represent the mobile station, base stations, controllers, switching center, and external networks.

4. Draw hexagonal cells and use different colors to show frequency groups.

5. Illustrate how a mobile device moves from one cell to another through the handover process.

6. Create a prototype mock-up showing cells, base stations, selected-cell information, signal strength, and handover status.

7. Check the design and export the frames for presentation.

Design

The five design figures below explain the main parts and operation of a basic cellular system.

1. Overall Architecture of a Cellular System

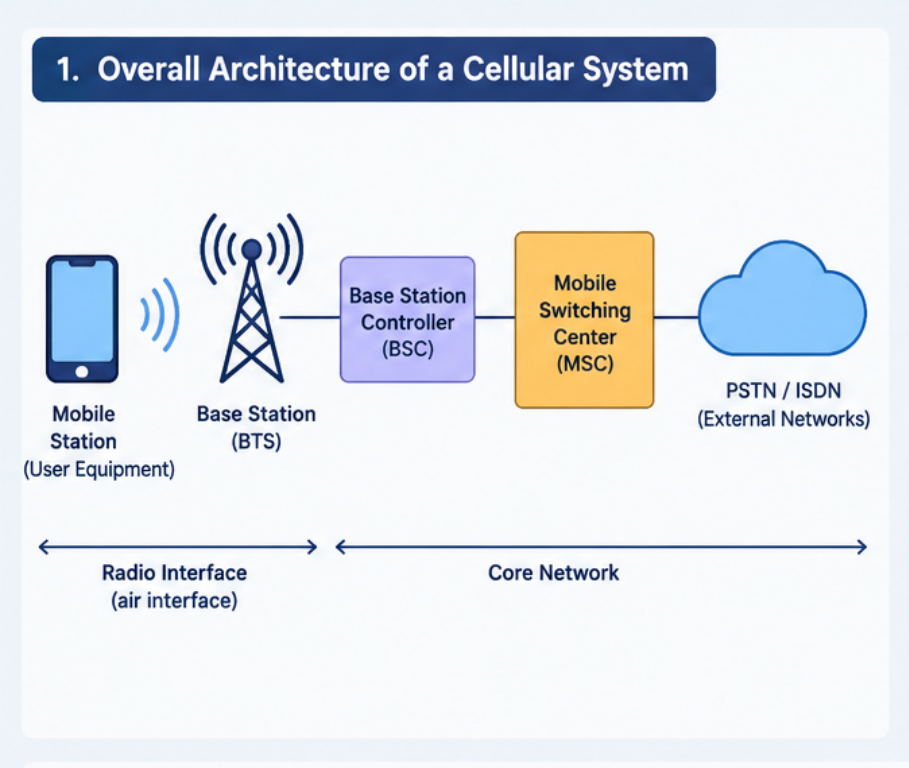


Figure 1: 1. Overall Architecture of a Cellular System.

2. Cellular Layout

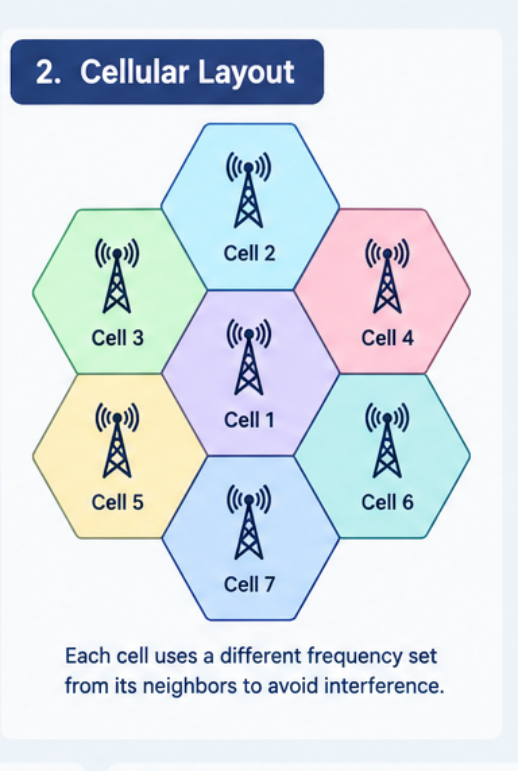


Figure 2: 2. Cellular Layout.

3. Frequency Reuse

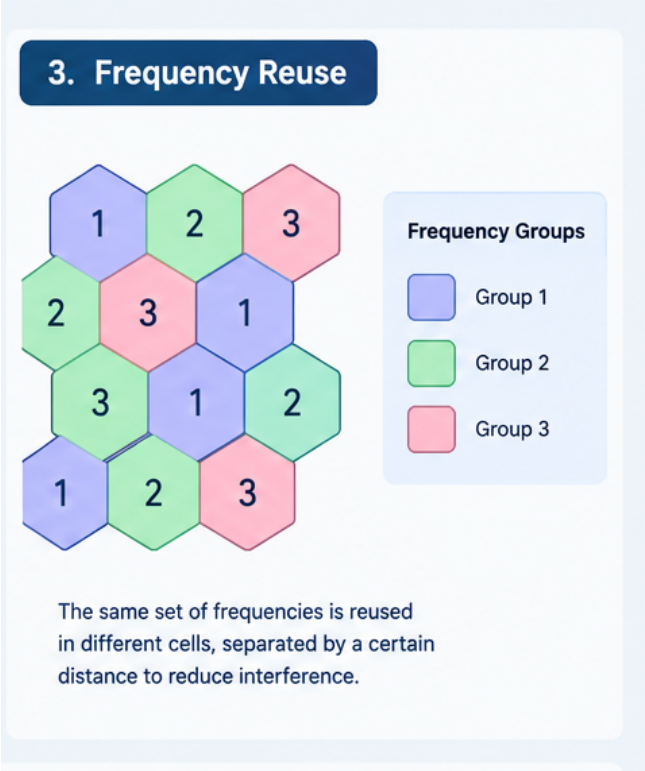


Figure 3: 3. Frequency Reuse.

4. Handover Process

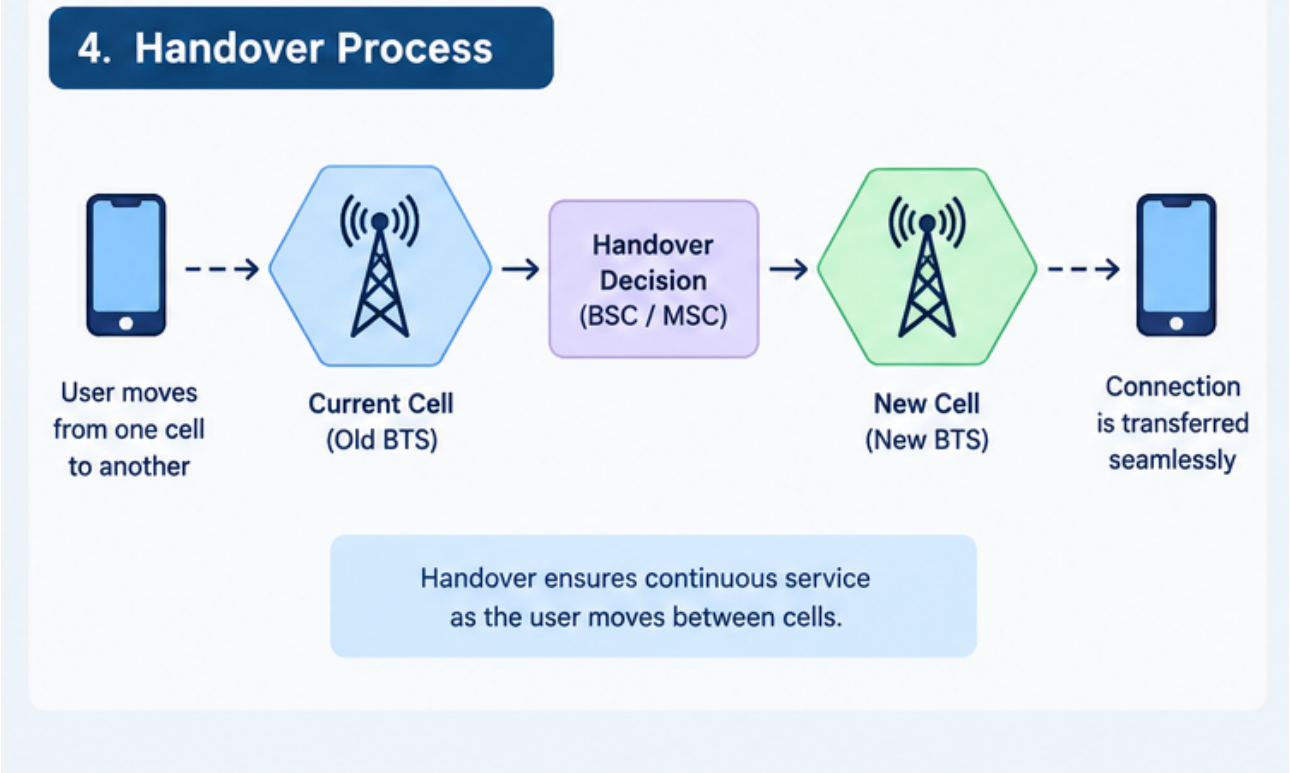


Figure 4: 4. Handover Process.

5. Key Components of a Cellular System

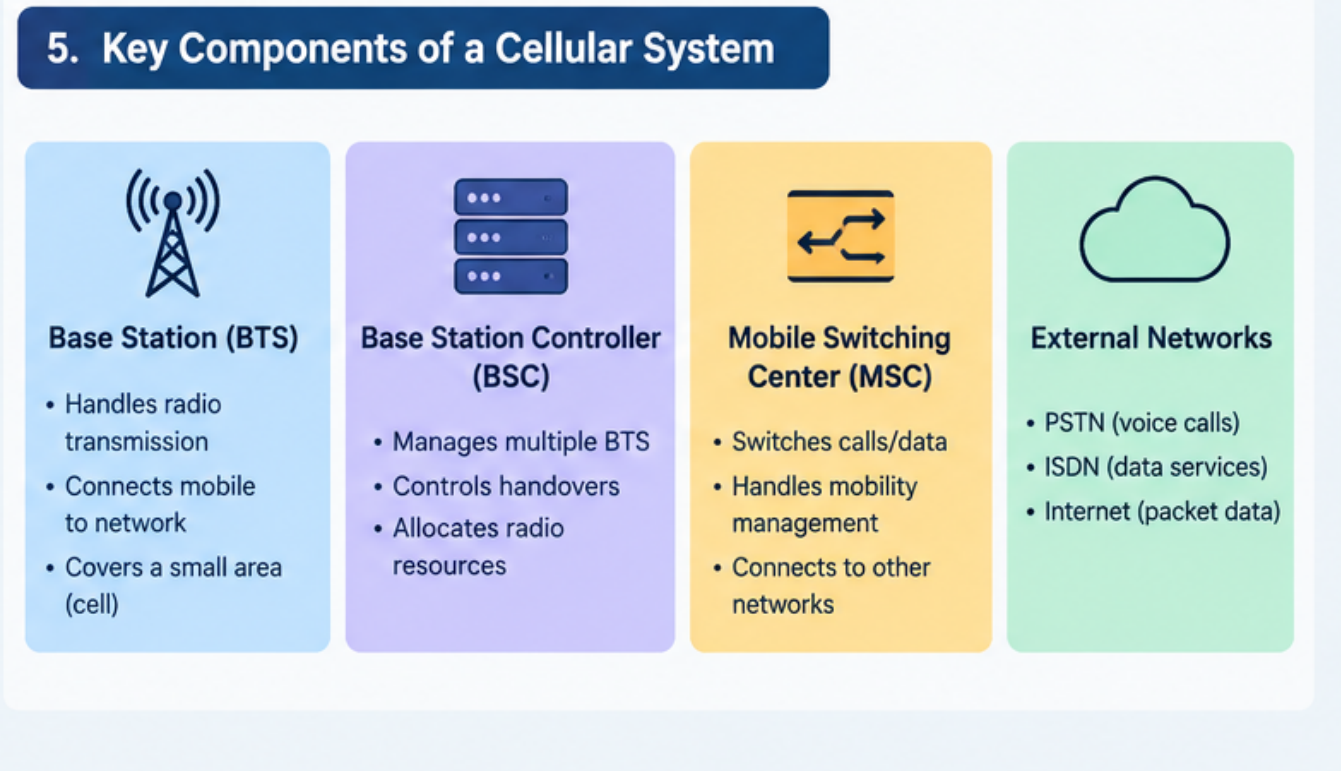


Figure 5: 5. Key Components of a Cellular System.

Prototype Design

The prototype is a visual mock-up of a simple cellular-network simulator. It shows the hexagonal cell layout, base stations, selected-cell information, network status, signal strength, and handover concept.

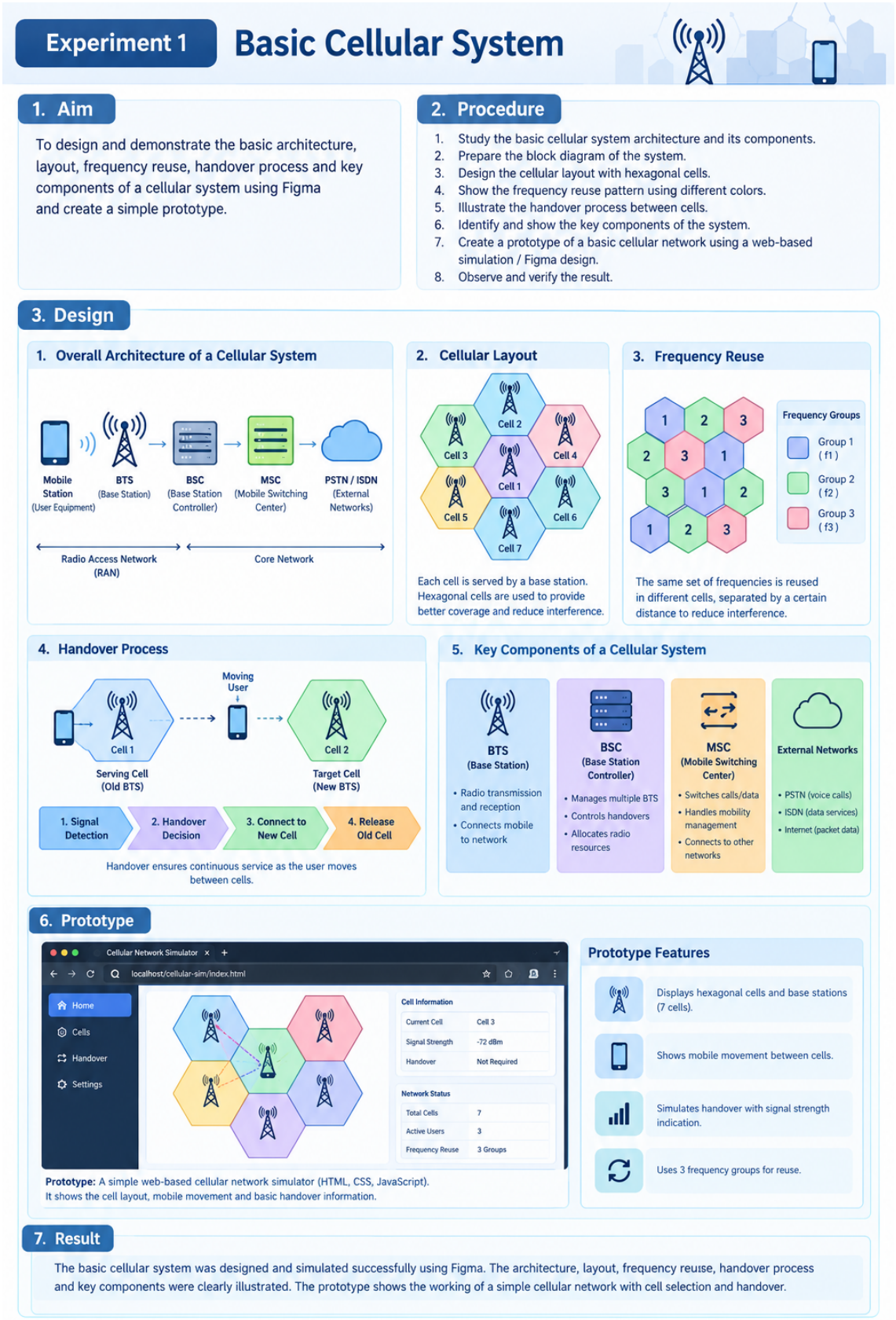


Figure 6: Prototype mock-up of a basic cellular network simulator.

Prototype features

- Displays seven hexagonal cells and their base stations.
- Shows selected-cell and network-status information.
- Illustrates mobile movement between cells and the handover concept.
- Uses signal-strength indicators and frequency reuse groups.

Result

The basic cellular system was designed using five Figma-style views, and a prototype mock-up was added to demonstrate cell selection, network status, frequency reuse, and handover. The designs illustrate the structure and basic operation of a cellular communication system.